

1. Explain the difference between Relative , Absolute and Mixed Cell Referencing.

Ans:-

Relative Cell Referencing :-

- a. Change automatically when copied to another cell.
- b. Example: If =A1 + B1 in cell C1 is copied to C2, it becomes =A2 + B2.
- c. Use case: Great for repeating calculations across rows or columns.
- d. ensures formulas adapt automatically.

Absolute Cell Referencing :-

- a. Remain fixed no matter where the formula is copied.
- b Written with \$ sign
→ \$A\$1.
- c Example: If =A1 * \$B\$1 is copied down, only A1 changes, \$B\$1 stays constant.
- d. Use case: Useful for fixed values like tax rates, conversion factors, constants.
- e. prevents errors when you always need the same value.

Mixed Cell Referencing:-

- a. Combination of relative and absolute.
- b. \$A1 → column A fixed, row changes.
- c. A\$1 → row 1 fixed, column changes.
- d. Use case: Helpful in tables or multiplying across both rows and columns.
- e. gives flexibility in row/column operations.

You have the following dataset:

Vehicle	Price
Car	500000
Bicycle	10500

Auto-Rickshaw	25000
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2. Write a formula to calculate the total sales of Car and Bicycle only.

Ans:-

D2				=PRODUCT(B2,C2)
	A	B	C	D
1	Vehicle	Price	Quantity	Total sales
2	Car	500000	10	5000000
3	Bicycle	10500	5	52500

3. Using the data below, write a formula to calculate the average sales of items priced above 100 but less than 300:

Ans:-

D8				=AVERAGEIFS(C8:C10,B8:B10,">100",B8:B10,"<300")		
	A	B	C	D	E	F
4						
5						
6						
7	Item	Price	Sales	Average Sales		
8	Item A	90	1000	1350		
9	Item B	150	1200			
10	Item C	250	1500			
11						

Ans:-

H2				=COUNTA(B2:B51)				
	A	B	C	D	E	F	G	H
1	OrderID	CustomerName	Region	Product	Quantity	UnitPrice	OrderDate	Total customer names
2	1001	Amit Sharma	North	Notebook	10	50	05-01-2024	50
3	1002	Neha Verma	South	Pen	20	10	06-01-2024	
4	1003	Rahul Singh	East	Marker	15	25	07-01-2024	
5	1004	Pooja Mehta	West	Notebook	5	50	08-01-2024	
6	1005	Ankit Jain	North	Pencil	30	5	10-01-2024	
7	1006	Sneha Patel	South	Notebook	12	50	11-01-2024	
8	1007	Vikas Gupta	East	Eraser	25	8	13-01-2024	
9	1008	Meera Nair	West	Pen	18	10	15-01-2024	
10	1009	Anjali Rao	North	Marker	22	25	17-01-2024	
11	1010	Manoj Das	South	Notebook	8	50	18-01-2024	
12	1011	Ritika Sen	East	Pencil	40	5	19-01-2024	
13	1012	Karan Malhotra	West	Eraser	35	8	20-01-2024	
14	1013	Aryan Joshi	North	Notebook	14	50	22-01-2024	
15	1014	Divya Kapoor	South	Pen	28	10	23-01-2024	
16	1015	Sahil Arora	East	Marker	12	25	24-01-2024	
17	1016	Muskan Jain	West	Pencil	25	5	26-01-2024	
18	1017	Rohan Gupta	North	Eraser	20	8	28-01-2024	
19	1018	Nidhi Sharma	South	Notebook	10	50	29-01-2024	
20	1019	Tarun Verma	East	Pen	15	10	30-01-2024	
21	1020	Simran Kaur	West	Marker	18	25	01-02-2024	
22	1021	Deepak Rao	North	Notebook	9	50	02-02-2024	
23	1022	Payal Mehta	South	Pencil	35	5	03-02-2024	
24	1023	Aditya Singh	East	Eraser	30	8	04-02-2024	
25	1024	Shreya Nair	West	Notebook	11	50	06-02-2024	
26	1025	Saniav Kumar	North	Pen	14	10	07-02-2024	

5. Calculate the Total Sales for each row using a formula.

Ans:-

H2								
	A	B	C	D	E	F	G	H
1	OrderID	CustomerName	Region	Product	Quantity	UnitPrice	OrderDate	Total Sales
2	1001	Amit Sharma	North	Notebook	10	50	05-01-2024	500
3	1002	Neha Verma	South	Pen	20	10	06-01-2024	200
4	1003	Rahul Singh	East	Marker	15	25	07-01-2024	375
5	1004	Pooja Mehta	West	Notebook	5	50	08-01-2024	250
6	1005	Ankit Jain	North	Pencil	30	5	10-01-2024	150
7	1006	Sneha Patel	South	Notebook	12	50	11-01-2024	600
8	1007	Vikas Gupta	East	Eraser	25	8	13-01-2024	200
9	1008	Meera Nair	West	Pen	18	10	15-01-2024	180
10	1009	Anjali Rao	North	Marker	22	25	17-01-2024	550
11	1010	Manoj Das	South	Notebook	8	50	18-01-2024	400
12	1011	Ritika Sen	East	Pencil	40	5	19-01-2024	200
13	1012	Karan Malhotra	West	Eraser	35	8	20-01-2024	280
14	1013	Aryan Joshi	North	Notebook	14	50	22-01-2024	700
15	1014	Divya Kapoor	South	Pen	28	10	23-01-2024	280
16	1015	Sahil Arora	East	Marker	12	25	24-01-2024	300
17	1016	Muskan Jain	West	Pencil	25	5	26-01-2024	125
18	1017	Rohan Gupta	North	Eraser	20	8	28-01-2024	160
19	1018	Nidhi Sharma	South	Notebook	10	50	29-01-2024	500
20	1019	Tarun Verma	East	Pen	15	10	30-01-2024	150
21	1020	Simran Kaur	West	Marker	18	25	01-02-2024	450
22	1021	Deepak Rao	North	Notebook	9	50	02-02-2024	450
23	1022	Payal Mehta	South	Pencil	35	5	03-02-2024	175
24	1023	Aditya Singh	East	Eraser	30	8	04-02-2024	240
25	1024	Shreya Nair	West	Notebook	11	50	06-02-2024	550
26	1025	Sanjay Kumar	North	Pen	14	10	07-02-2024	140
27	1026	Aarti Sharma	South	Marker	16	25	08-02-2024	400
28	1027	Rajesh Patel	East	Pencil	22	5	09-02-2024	110
29	1028	Preeti Joshi	West	Eraser	27	8	10-02-2024	216

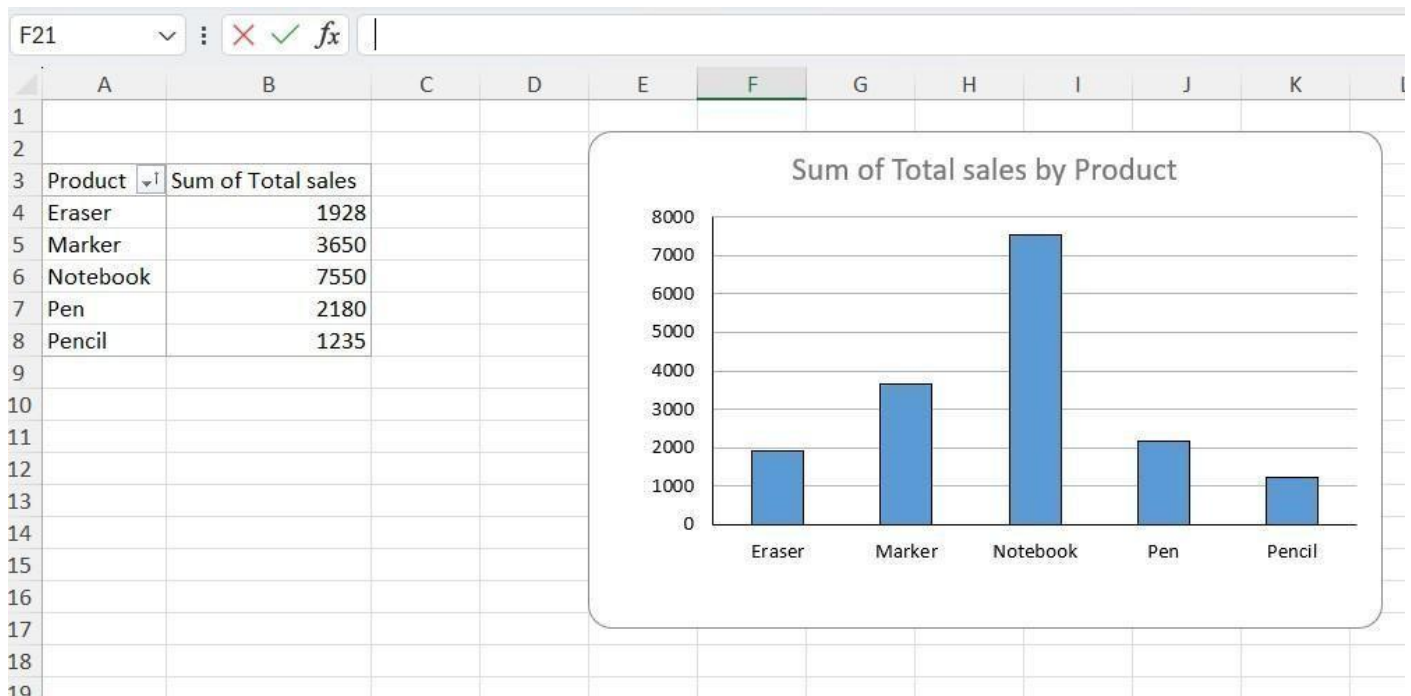
6. Calculate the total sales of Notebooks in the North region only.

Ans:-

H2								{=SUMPRODUCT((C2:C51=C2)*(D2:D51=D2)*(E2:E51)*(F2:F51))}
	A	B	C	D	E	F	G	H
1	OrderID	CustomerName	Region	Product	Quantity	UnitPrice	OrderDate	total sales of Notebooks
2	1001	Amit Sharma	North	Notebook	10	50	05-01-2024	2900
3	1002	Neha Verma	South	Pen	20	10	06-01-2024	
4	1003	Rahul Singh	East	Marker	15	25	07-01-2024	
5	1004	Pooja Mehta	West	Notebook	5	50	08-01-2024	
6	1005	Ankit Jain	North	Pencil	30	5	10-01-2024	
7	1006	Sneha Patel	South	Notebook	12	50	11-01-2024	
8	1007	Vikas Gupta	East	Eraser	25	8	13-01-2024	
9	1008	Meera Nair	West	Pen	18	10	15-01-2024	
10	1009	Anjali Rao	North	Marker	22	25	17-01-2024	
11	1010	Manoj Das	South	Notebook	8	50	18-01-2024	
12	1011	Ritika Sen	East	Pencil	40	5	19-01-2024	
13	1012	Karan Malhotra	West	Eraser	35	8	20-01-2024	
14	1013	Aryan Joshi	North	Notebook	14	50	22-01-2024	
15	1014	Divya Kapoor	South	Pen	28	10	23-01-2024	
16	1015	Sahil Arora	East	Marker	12	25	24-01-2024	
17	1016	Muskan Jain	West	Pencil	25	5	26-01-2024	
18	1017	Rohan Gupta	North	Eraser	20	8	28-01-2024	
19	1018	Nidhi Sharma	South	Notebook	10	50	29-01-2024	
20	1019	Tarun Verma	East	Pen	15	10	30-01-2024	
21	1020	Simran Kaur	West	Marker	18	25	01-02-2024	
22	1021	Deepak Rao	North	Notebook	9	50	02-02-2024	
23	1022	Payal Mehta	South	Pencil	35	5	03-02-2024	
24	1023	Aditya Singh	East	Eraser	30	8	04-02-2024	
25	1024	Shreya Nair	West	Notebook	11	50	06-02-2024	
26	1025	Sanjay Kumar	North	Pen	14	10	07-02-2024	
27	1026	Aarti Sharma	South	Marker	16	25	08-02-2024	
28	1027	Rajesh Patel	East	Pencil	22	5	09-02-2024	
29	1028	Preeti Joshi	West	Eraser	27	8	10-02-2024	

7. Create a column chart showing total sales by product.

Ans:-



8. Insert a line chart showing daily sales trend.

Ans:-

